

INDUSTRIAL ENGINEERING & ASSET RELIABILITY REPORT

Synthetic maintenance and manufacturing dataset designed to test structured extraction, nested tables, measurements, units, dates, alarm codes, and technical terminology.

Plant	Pune Manufacturing Campus	Reporting Window	01 Apr – 30 Jun 2026
Assets Monitored	184	Production Hours	21,480 h
Unplanned Downtime	318.5 h	Availability	98.52%
Critical Alarms	47	MTBF	426.8 h

1. Plant Performance

Line	Product Family	Units	Yield	Scrap	Downtime	OEE	Status
L-01	Sensor Pack	17,625	98.58%	1.42%	10.2 h	80.38%	Watch
L-02	Control Module	27,381	97.85%	2.15%	30.7 h	92.53%	Nominal
L-03	Gearbox	11,590	96.46%	3.54%	18.0 h	85.62%	Nominal
L-04	Gearbox	10,275	98.59%	1.41%	37.9 h	79.77%	Watch
L-05	Sensor Pack	19,618	99.06%	0.94%	6.4 h	81.45%	Watch
L-06	Servo Motor	23,200	96.62%	3.38%	9.1 h	86.33%	Nominal
L-07	Drive Unit	23,020	99.09%	0.91%	33.3 h	79.76%	Watch
L-08	Control Module	26,213	99.19%	0.81%	37.4 h	94.67%	Nominal
L-09	Servo Motor	13,949	97.60%	2.40%	19.8 h	93.56%	Nominal
L-10	Gearbox	13,894	97.50%	2.50%	38.0 h	82.77%	Watch
L-11	Sensor Pack	17,847	97.67%	2.33%	37.0 h	95.92%	Nominal
L-12	Sensor Pack	24,447	96.67%	3.33%	13.2 h	95.44%	Nominal

2. Asset Register & Sensor Telemetry

Asset ID	Equipment	Serial	Temp °C	Vibration mm/s	Pressure bar	RPM	Current A	Health
A-0001	Robotic Arm	SN26-159642	71.5	1.33	8.11	830	29.79	Watch
A-0002	Compressor	SN26-523389	83.8	1.50	6.23	5,294	7.18	Critical
A-0003	Hydraulic Press	SN26-539589	37.6	8.55	7.12	5,290	5.94	Critical
A-0004	Laser Cutter	SN26-350280	67.2	5.72	8.15	2,556	13.15	Healthy
A-0005	Hydraulic Press	SN26-109767	45.6	1.94	12.99	4,165	18.05	Healthy
A-0006	Conveyor Drive	SN26-238890	56.2	9.49	18.14	1,020	27.75	Critical
A-0007	Test Chamber	SN26-974244	82.3	8.46	16.43	3,447	16.68	Critical
A-0008	Conveyor Drive	SN26-795613	60.9	3.43	17.96	5,777	27.37	Healthy
A-0009	Robotic Arm	SN26-385577	36.7	8.44	7.39	1,296	36.72	Critical
A-0010	Conveyor Drive	SN26-629959	46.5	2.63	8.54	6,051	31.98	Healthy
A-0011	Conveyor Drive	SN26-146228	57.9	8.68	15.94	1,176	31.98	Critical
A-0012	Test Chamber	SN26-678478	31.2	7.51	12.87	2,565	11.28	Watch
A-0013	CNC-5 Axis	SN26-975136	69.8	5.68	5.14	6,080	43.79	Healthy
A-0014	CNC-5 Axis	SN26-475190	51.3	5.61	9.82	762	17.70	Healthy
A-0015	Injection Molder	SN26-750810	42.6	2.97	5.01	5,006	42.93	Healthy

Asset ID	Equipment	Serial	Temp °C	Vibration mm/s	Pressure bar	RPM	Current A	Health
A-0016	Robotic Arm	SN26-532322	72.2	8.85	6.98	1,751	46.96	Critical
A-0017	Compressor	SN26-379766	32.4	7.21	8.38	3,792	39.34	Watch
A-0018	Compressor	SN26-309267	39.8	6.90	9.12	737	41.81	Watch
A-0019	Compressor	SN26-517821	75.9	4.80	7.99	2,284	13.96	Watch
A-0020	Injection Molder	SN26-812526	49.1	8.32	17.71	2,706	19.58	Critical
A-0021	Conveyor Drive	SN26-287241	85.0	5.47	17.31	1,364	42.61	Critical
A-0022	Laser Cutter	SN26-557592	62.9	9.36	4.06	5,307	23.23	Critical
A-0023	CNC-5 Axis	SN26-843215	64.3	5.25	9.16	5,143	12.53	Healthy
A-0024	Compressor	SN26-481913	55.0	5.32	15.36	6,046	35.71	Healthy

3. Maintenance Work Orders

WO	Asset	Opened	Priority	Failure Mode	Action	Labor h	Parts ■	SLA
WO-26-0001	A-0003	26/05/2026	P3	Motor overload	Clean & recalibrate	13.1	■79,527	24h
WO-26-0002	A-0014	25/06/2026	P3	PLC communication	Align and tension	10.2	■51,089	72h
WO-26-0003	A-0013	25/06/2026	P2	Belt misalignment	Replace module	10.1	■906	24h
WO-26-0004	A-0007	07/05/2026	P4	Motor overload	Flush system	8.2	■177,911	8h
WO-26-0005	A-0016	05/06/2026	P2	Overtemperature	Align and tension	9.5	■166,721	24h
WO-26-0006	A-0008	12/04/2026	P3	Sensor drift	Firmware update	3.1	■12,914	8h
WO-26-0007	A-0020	31/05/2026	P1	Coolant contamination	Replace module	16.0	■151,708	8h
WO-26-0008	A-0013	29/06/2026	P4	PLC communication	Firmware update	3.1	■181,056	4h
WO-26-0009	A-0014	14/04/2026	P2	Seal leakage	Flush system	1.4	■66,125	4h
WO-26-0010	A-0005	29/05/2026	P4	Motor overload	Flush system	11.2	■133,124	72h
WO-26-0011	A-0015	10/06/2026	P2	Coolant contamination	Flush system	5.0	■65,612	24h
WO-26-0012	A-0016	06/06/2026	P2	Belt misalignment	Flush system	1.9	■75,701	8h
WO-26-0013	A-0011	05/05/2026	P3	Overtemperature	Inspect and reseal	3.1	■101,210	8h
WO-26-0014	A-0007	30/06/2026	P1	PLC communication	Replace module	6.3	■122,939	72h
WO-26-0015	A-0007	08/04/2026	P4	PLC communication	Replace component	15.5	■151,713	72h
WO-26-0016	A-0001	01/06/2026	P3	Belt misalignment	Replace module	15.4	■110,643	8h
WO-26-0017	A-0008	02/06/2026	P3	PLC communication	Flush system	1.0	■88,914	72h
WO-26-0018	A-0015	22/04/2026	P2	Bearing wear	Replace module	10.9	■174,601	4h
WO-26-0019	A-0021	11/04/2026	P4	Seal leakage	Flush system	3.7	■68,998	72h
WO-26-0020	A-0007	12/05/2026	P4	Motor overload	Electrical inspection	13.8	■100,185	24h
WO-26-0021	A-0009	24/05/2026	P1	Coolant contamination	Replace component	13.6	■14,453	24h
WO-26-0022	A-0021	29/04/2026	P1	Bearing wear	Replace component	17.1	■53,061	4h
WO-26-0023	A-0005	19/06/2026	P2	Seal leakage	Flush system	12.2	■148,641	8h
WO-26-0024	A-0023	30/05/2026	P3	Motor overload	Inspect and reseal	11.1	■30,824	8h
WO-26-0025	A-0004	10/05/2026	P1	Belt misalignment	Replace module	7.4	■52,790	4h
WO-26-0026	A-0023	15/06/2026	P2	Overtemperature	Align and tension	15.4	■158,194	4h
WO-26-0027	A-0002	12/06/2026	P3	PLC communication	Electrical inspection	1.7	■170,549	24h
WO-26-0028	A-0014	02/04/2026	P4	Overtemperature	Replace module	17.3	■167,414	72h
WO-26-0029	A-0005	30/06/2026	P4	Seal leakage	Align and tension	11.3	■141,878	72h
WO-26-0030	A-0014	30/05/2026	P3	Motor overload	Firmware update	15.0	■23,518	24h
WO-26-0031	A-0008	28/05/2026	P4	PLC communication	Electrical inspection	1.0	■86,000	8h
WO-26-0032	A-0007	02/06/2026	P3	Belt misalignment	Electrical inspection	5.4	■157,078	24h
WO-26-0033	A-0001	11/06/2026	P2	Overtemperature	Firmware update	13.1	■128,877	8h
WO-26-0034	A-0016	28/06/2026	P4	Coolant contamination	Replace component	2.1	■58,890	72h
WO-26-0035	A-0008	28/06/2026	P3	Motor overload	Flush system	10.2	■90,913	72h
WO-26-0036	A-0011	10/06/2026	P3	Coolant contamination	Align and tension	5.9	■61,234	4h
WO-26-0037	A-0011	25/04/2026	P1	Seal leakage	Firmware update	4.3	■127,729	24h
WO-26-0038	A-0017	15/06/2026	P3	Overtemperature	Firmware update	5.7	■95,402	8h
WO-26-0039	A-0001	09/05/2026	P2	Belt misalignment	Replace component	17.6	■145,859	24h

WO	Asset	Opened	Priority	Failure Mode	Action	Labor h	Parts ■	SLA
WO-26-0040	A-0005	29/06/2026	P4	Overtemperature	Replace component	10.5	■123,849	72h

4. Alarm Codes & Diagnostic Notes

Code	Meaning	Trigger	Recommended Response
E-104	High spindle temperature	T > 78 °C for 120 s	Reduce load; inspect coolant circuit
V-207	Excess vibration	RMS > 7.5 mm/s	Check bearing, coupling and alignment
P-301	Low hydraulic pressure	P < 4.0 bar	Inspect pump, filter and leakage points
C-412	Drive current spike	I > rated current × 1.35	Inspect mechanical load and inverter
N-515	Network packet loss	Loss > 2.0% / 60 s	Check industrial Ethernet path

5. Engineering Notes

Measurement precision is intentionally mixed: 78.0 °C, 7.500 mm/s, 4.00 bar, 1.35× rated current, and 2.0%. The document also includes unit-bearing values, serial numbers, hyphenated identifiers, and status transitions to test parsers that depend on punctuation and formatting.